

UAS TEORI

2. Support Vector Machine (SVM) is a set of supervised learning methods used for classification, regression, and outlier detection. **[Total: 30 points]**

a) What makes SVM different with other machine learning models? **[5 points]**

b) Perform dataset creation as instructed in the code below: **[2 Points]**

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

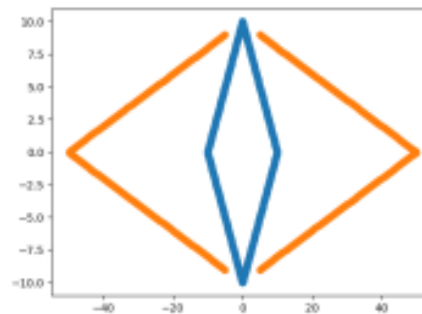
def rhombus_equation(x, a):
    return a - np.abs(x)

#Membuat dataset
x = np.linspace(-10.0, 10.0, 200)
a_value1 = 10
y = rhombus_equation(x, a_value1)
y=np.hstack([y,-y])
x=np.hstack([x,-x])

x1 = np.linspace(-9.0, 9.0, 200)
a_value2 = 50
y1 = rhombus_equation(5*x1, a_value2)
y1=np.hstack([y1,-y1])
x1=np.hstack([x1,-x1])

df1 =pd.DataFrame(np.vstack([y,x]).T,columns=['X1','X2'])
df1['Y']=0
df2 =pd.DataFrame(np.vstack([y1,x1]).T,columns=['X1','X2'])
df2['Y']=1
# Menggabungkan DataFrames menggunakan pd.concat
df = pd.concat([df1, df2], ignore_index=True)
df.head()
```

```
#Run in different cell
plt.scatter(y,x)
plt.scatter(y1,x1)
```



- c) Using the dataset you created before, perform training and testing using **two** different tuning sets of **linear** kernel SVM models. Train and test ratio is 80:20. [7 points]
- d) Using the dataset you created before, perform training and testing using two different tuning sets of **Radial Basis Function (RBF)** SVM model. Train and test ratio is 80:20. [7 points]
- e) Based on experiments 2c and 2d, complete the following table: [4 points]

SDM model	Differences Parameters
SVM model with linear kernel type 1	
SVM model with linear kernel type 2	
SVM model with RBF kernel type 1	
SVM model with RBF kernel type 2	

Table 2.1. The differences parameter used between SVM models.

	Metrics	SVM Model Evaluation using testing results			
		with Linear Kernel type 1	with Linear Kernel type 2	with RBF Kernel type 1	with RBF Kernel type 2
Overall	Accuracy				
	Support				
Class 0	Precision				
	Recall				
	f1-Score				
	Support				
Class 1	Precision				
	Recall				
	f1-Score				
	Support				

Table 2.2. Evaluation metrics between all SVM models.

- f) Perform analysis based on the obtained results in Table 2.2. (Suggestion: discuss which one is the best, then why, and which one is the worst, then why.) [5 points]

3. The idea of **ensemble learning** is to build a prediction model by combining the strengths of a collection of simpler base models, called weak learners. Bagging is one of ensemble learning. In this question, we will compare a single decision tree model with a bagging model containing 100 decision trees. The **HeartFailurePredictionDataset.csv** dataset used can be seen below. **[Total: 30 points]**

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	40	M	ATA	140	289	0	Normal	172	N	0.0	Up	0
1	49	F	NAP	160	180	0	Normal	156	N	1.0	Flat	1
2	37	M	ATA	130	283	0	ST	98	N	0.0	Up	0
3	48	F	ASY	138	214	0	Normal	108	Y	1.5	Flat	1
4	54	M	NAP	150	195	0	Normal	122	N	0.0	Up	0
...	...	...	...	...	...	...	...	...	...	...	...	...
913	45	M	TA	110	264	0	Normal	132	N	1.2	Flat	1
914	68	M	ASY	144	193	1	Normal	141	N	3.4	Flat	1
915	57	M	ASY	130	131	0	Normal	115	Y	1.2	Flat	1
916	57	F	ATA	130	236	0	LVH	174	N	0.0	Flat	1
917	38	M	NAP	138	175	0	Normal	173	N	0.0	Up	0

918 rows x 12 columns

Attribute Information

- Age: age of the patient [years]
- Sex: sex of the patient [M: Male, F: Female]
- ChestPainType: chest pain type [TA: Typical Angina, ATA: Atypical Angina, NAP: Non-Anginal Pain, ASY: Asymptomatic]
- RestingBP: resting blood pressure [mm Hg]
- Cholesterol: serum cholesterol [mm/dl]
- FastingBS: fasting blood sugar [1: if FastingBS > 120 mg/dl, 0: otherwise]
- RestingECG: resting electrocardiogram results [Normal: Normal, ST: having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV), LVH: showing probable or definite left ventricular hypertrophy by Estes' criteria]
- MaxHR: maximum heart rate achieved [Numeric value between 60 and 202]
- ExerciseAngina: exercise-induced angina [Y: Yes, N: No]
- Oldpeak: oldpeak = ST [Numeric value measured in depression]
- ST_Slope: the slope of the peak exercise ST segment [Up: upsloping, Flat: flat, Down: downsloping]
- HeartDisease: output class [1: heart disease, 0: Normal]

- Perform feature engineering into our dataset. **[5 points]**
- Using the HeartFailurePredictionDataset dataset, perform training and testing using a single decision tree model. Train and test ratio is 80:20. **[8 points]**
- Using the HeartFailurePredictionDataset dataset, perform training and testing using the Bagging model, which contains 100 decision tree models. Train and test ratio is 80:20. **[8points]**

- d) Based on experiments 3b and 3c, complete the following table: **[2 points]**

	Metrics	Evaluation	
		single Decision Tree model	Bagging model containing 100 decision tree models
Overall	Accuracy		
	Support		
Class 0	Precision		
	Recall		
	f1-Score		
	Support		
Class 1	Precision		
	Recall		
	f1-Score		
	Support		

Table 3.1. Evaluation metrics comparison between two different models.

- e) Perform analysis based on the obtained results in Table 3.1. (Suggestion: discuss which one is the best, then why, and which one is the worst, then why.) **[7 points]**

UAS LAB

1. You work as a data scientist at an international-level institution "Justice League". You are asked to dig up information in a dataset, namely “**World Educational Data**” by looking for data that has the same characteristics and then collecting it into several groups based on the similarity of the data. [100%]
 - a) As a data scientist, what machine learning approach will you use to achieve this goal?
 - b) Implement three different models to achieve that goal.

- c) Modify the following visualization code to display the results of the two different models that you have done. (You can choose any three features from the dataset; you have to visualize all the model results you performed.)

```
import matplotlib.pyplot as plt

# Create a 3D plot
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')

# Scatter plot with colors based on 'k_means' column
scatter = ax.scatter(df['feat1'], df['feat2'], df['feat3'],
c=df['models_result'], cmap='viridis')

# Add color bar
color_bar = fig.colorbar(scatter, ax=ax)
color_bar.set_label('model_name')

# Set axis labels
ax.set_xlabel('feat1')
ax.set_ylabel('feat2')
ax.set_zlabel('feat3')

# Show the plot
plt.show()
```

- d) Provide an analysis of the comparison of the results you got.

Dataset used:**World Educational Data****Description:**

This meticulously curated dataset offers a **panoramic view of education on a global scale**, delivering profound insights into the dynamic landscape of education across diverse countries and regions. Spanning a rich tapestry of educational aspects, it encapsulates crucial metrics including out-of-school rates, completion rates, proficiency levels, literacy rates, birth rates, and primary and tertiary education enrollment statistics. A treasure trove of knowledge, this dataset is an indispensable asset for discerning researchers, dedicated educators, and forward-thinking policymakers, enabling them to embark on a transformative journey of assessing, enhancing, and reshaping education systems worldwide.

1. **Countries and Areas:** Name of the countries and areas.
2. **Latitude:** Latitude coordinates of the geographical location.
3. **Longitude:** Longitude coordinates of the geographical location.
4. **OOSR_Pre0Primary_Age_Male:** Out-of-school rate for pre-primary age males.
5. **OOSR_Pre0Primary_Age_Female:** Out-of-school rate for pre-primary age females.
6. **OOSR_Primary_Age_Male:** Out-of-school rate for primary age males.
7. **OOSR_Primary_Age_Female:** Out-of-school rate for primary age females.
8. **OOSR_Lower_Secondary_Age_Male:** Out-of-school rate for lower secondary age males.

9. **OOSR_Lower_Secondary_Age_Female**: Out-of-school rate for lower secondary age females.
10. **OOSR_Upper_Secondary_Age_Male**: Out-of-school rate for upper secondary age males.
11. **OOSR_Upper_Secondary_Age_Female**: Out-of-school rate for upper secondary age females.
12. **Completion_Rate_Primary_Male**: Completion rate for primary education among males.
13. **Completion_Rate_Primary_Female**: Completion rate for primary education among females.
14. **Completion_Rate_Lower_Secondary_Male**: Completion rate for lower secondary education among males.
15. **Completion_Rate_Lower_Secondary_Female**: Completion rate for lower secondary education among females.
16. **Completion_Rate_Upper_Secondary_Male**: Completion rate for upper secondary education among males.
17. **Completion_Rate_Upper_Secondary_Female**: Completion rate for upper secondary education among females.
18. **Grade_2_3_Proficiency_Reading**: Proficiency in reading for grade 2-3 students.
19. **Grade_2_3_Proficiency_Math**: Proficiency in math for grade 2-3 students.
20. **Primary_End_Proficiency_Reading**: Proficiency in reading at the end of primary education.
21. **Primary_End_Proficiency_Math**: Proficiency in math at the end of primary education.
22. **Lower_Secondary_End_Proficiency_Reading**: Proficiency in reading at the end of lower secondary education.
23. **Lower_Secondary_End_Proficiency_Math**: Proficiency in math at the end of lower secondary education.
24. **Youth_15_24_Literacy_Rate_Male**: Literacy rate among male youths aged 15-24.
25. **Youth_15_24_Literacy_Rate_Female**: Literacy rate among female youths aged 15-24.
26. **Birth_Rate**: Birth rate in the respective countries/areas.
27. **Gross_Primary_Education_Enrollment**: Gross enrollment in primary education.
28. **Gross_Tertiary_Education_Enrollment**: Gross enrollment in tertiary education.
29. **Unemployment_Rate**: Unemployment rate in the respective countries/areas.